

Note: Write all the required equations (without simplification) to solve the following problems, then find the final answer.

Prob.1:

Consider the circuit shown in figure 1, where a voltage source $v_s(t) = 100 \cos(314t + 20^\circ) V$ supplies the load by a current $i(t) = 10 \cos(314t - 10^\circ) A$

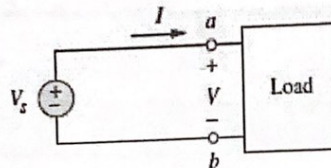
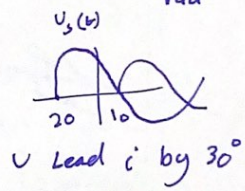


Fig.1

1. The equivalent impedance at the terminals of the voltage source is $Z = \dots \Omega$
2. The voltage leads the current by $\dots 30^\circ \dots$ degrees.
3. The current lags the voltage by $\dots \frac{\pi}{6} \dots \Rightarrow 0.524 \dots$ radians.
4. The voltage value $v_s(t)$ at $t = 5ms$ is $\dots 34.13 \dots V$
5. The phasor domain representation of $i(t)$ is $10 \angle -10^\circ A$
6. The frequency of the voltage $v_s(t)$ in Hz is $\dots 499.7 \dots Hz$.
7. Draw the phasor diagram for the voltage $v_s(t)$ and the current $i_s(t)$.

$$10 \angle 30^\circ \Rightarrow 8.66 + 5j$$

$$\boxed{3} \quad \frac{30}{\text{deg}} \times \frac{\pi}{180} = \frac{1}{6} \pi \text{ rad}$$

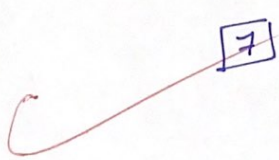


v Lead i by 30°

$$\dot{V}_s = 100 \angle 20^\circ \quad \dot{I}_s = 10 \angle -10^\circ$$

$$\boxed{1} \quad Z_{eq} = \frac{V_s}{I_s} = 10 \angle 30^\circ = 8.66 + 5j$$

$$\boxed{4} \quad 100 \cos\left(314 \times 5 \times 10^{-3} \times \frac{180}{\pi} + 20^\circ\right) = 34.127 \text{ V}$$



Prob.2: Consider the circuit shown in figure 2,

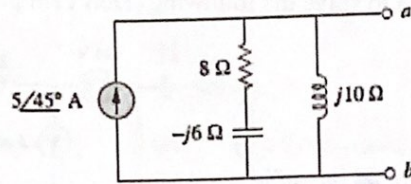
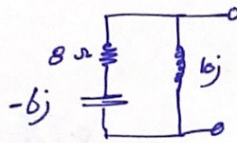


Fig. 2

a. Find the Thevenin equivalent impedance between the a - b terminals.

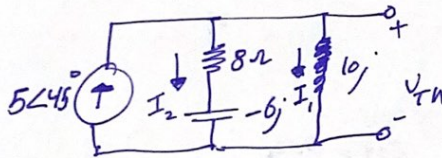


$$(8 - j6) \parallel j10 = \frac{(8 - j6)(j10)}{8 - j6 + j10}$$

$$Z_{Th} = 10 + 5j \quad \Omega$$

$$Z_{Th} = 11.2 \angle 26.5^\circ$$

b. Find the Thevenin voltage between the a - b terminals.



$$V_{Th} = V_{on}(j10)$$

$$V_{on}(j10) = j10(I_1)$$

$$I_1 = (5 \angle 45^\circ) \left(\frac{8 - j6}{8 - j6 + j10} \right)$$

$$V_{Th} = 5.59 \angle -18.4^\circ \text{ V}$$

$$V_{Th} = 5.30 - j1.76$$

Prob. 3: Consider the circuit shown in figure 3. Using superposition theorem, write the required equation(s) that you need to solve the following (Don't simplify)

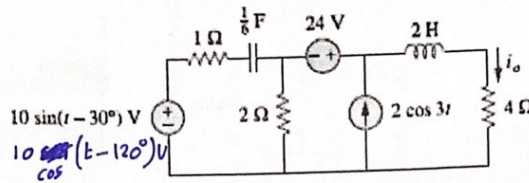


Fig. 3

a. The current i_o' due to the current source only $\omega = 3$

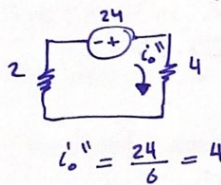
$$(1-2j) \parallel 2 = \frac{14}{13} - \frac{8}{13}j$$

$$6j+4 = Z_{\text{right}}$$

$$i_o' = I_s \left(\frac{\frac{14}{13} - \frac{8}{13}j}{\frac{14}{13} - \frac{8}{13}j + 6j + 4} \right) = 0.335 \angle -76.43^\circ$$

$$i_o'(t) = 0.335 \cos(3t - 76.43^\circ) \text{ A}$$

b. The current i_o'' due to DC voltage source only



$$i_o''(t) = 4 \text{ A}$$

c. The current i_o''' due to AC voltage source only

$$[(2j+4) \parallel 2] + (1-j) = Z_{\text{eq}} = 9-2j$$

$$i_o''' = I_s \left(\frac{2}{2+2j+4} \right) = 0.343 \angle -125.9^\circ$$

$$i_o'''(t) = 0.343 \cos(t - 125.9^\circ) \text{ A}$$

Prob. 4: Using the nodal method, write the required equations that you need to find V_1 , V_2 , and V_3 , in circuit shown in figure 4.

super node (1,2):

$$\frac{V_1}{2} + \frac{V_1 - V_3}{2j} + \frac{V_2 - V_3}{-4j} = 0$$

$$\frac{V_1 - V_3}{2} + \frac{V_2 - V_3}{4} = 0 \quad [1]$$

node (3):

$$\frac{V_2 - V_3}{4} + \frac{V_1 - V_3}{2j} + 0.2V_1 = 0$$

$$\frac{V_2 - V_3}{4} + \frac{V_1 - V_3}{2j} + 0.2V_1 = 0 \quad [2]$$

Good Luck.

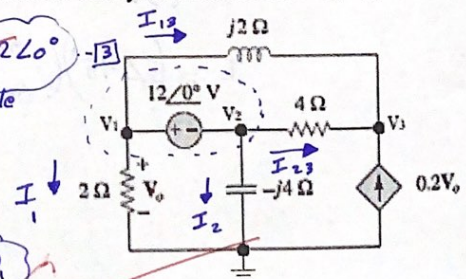


Fig. 4

$$V_1 = 7.68 \angle 50.2^\circ$$

$$V_2 = 9.219 \angle 140.2^\circ$$

$$V_3 = 3.887 \angle 68.63^\circ$$

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